

Wei-Jhih Huang

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Research Interests

AI-Native Wireless Networks and Next-Generation Radio Access Networks, Deep Learning for Cross-Domain Applications, Learning-Based Cross-Layer Resource Allocation and Optimization, Adaptive Learning with Out-of-Distribution Detection

Education

Bachelor, Communication Engineering, National Central University

Aug 2022 – Present

GPA: 3.94/4.3, Expected Graduation: July 2026

Taoyuan, Taiwan

Relevant Courses: Linear Algebra, Data Structures, Algorithms, Signal and System, Communication Theory, Digital Signal Processing, Digital Image Processing, Computer Organization, Digital System Design, Microcomputers, Computer Networks, and Programming for Deep Learning

Research Experience

Information Processing and Communications Lab, National Central University

Feb 2024 – Present

Undergraduate Research Student, Advisor: **Prof. Chih-Wei Huang**

Taoyuan, Taiwan

Keywords: Wireless Networking, Resource Allocation, Deep Reinforcement Learning, Generative models

- Designed a learning-based RRM framework improving convergence and robustness for dynamic resource orchestration in high-mobility B5G systems, achieving **67%** faster convergence under dynamic user trajectories.
- Built a trajectory-embedding mechanism for scenario distribution-aware DRL fine-tuning, reducing unnecessary DRL retraining by **90%** and improving user quality of service by **11%** in cross-layer resource management scenarios.

Electronic Design Automation Lab, The Chinese University of Hong Kong

Jul – Aug 2025

Summer Visiting Research Student, Advisor: **Prof. Bei Yu**

Hong Kong

Keywords: Electronic Design Automation, Deep Graph Learning, Timing Prediction, Logic synthesis

- Enhanced "**NUA-Timer++**" for graph-based hierarchical circuit timing prediction under Non-Uniform Input Arrival Times, achieving a **1.96x** improvement in consistency and reducing average error to **0.42x** compared to SOTA.
- Co-authored IEEE TCAD (Top-tier journal) submission as second author, collaborating with a Postdoc to leverage my machine learning expertise in Transformer architecture and loss function refinement for **NUA-Timer++** optimization.

Publications

- **Wei-Jhih Huang**, Jin-Min Yang, Ning Ma, Sheng-Liang Wu, Chih-Wei Huang. "Deviation-Aware Trajectory Embedding for Fine-Tuning Triggering in DRL-Based Wireless Network Resource Management," *IEEE Wireless Communications and Networking Conference (WCNC)*, 2026. (Accepted)
- Ziyi Wang, **Wei-Jhih Huang**, Fangzhou Liu, Tsung-Yi Ho, David Z. Pan, Bei Yu. "NUA-Timer++: Timing Prediction for Hierarchical RTL under Non-Uniform Input Arrival Times," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, under review.

Honors & Awards

- **Undergraduate Research Fellowship**, National Science and Technology Council (NSTC), Taiwan, 2025–2026
- **Honorable Mention**, Undergraduate Project Competition, Department of Communication Engineering, NCU, 2025
- **Merit Award**, Individual Service-Learning Excellence Scholarship, National Central University, 2025
- **Excellence Award**, NCU Club Evaluation (as Presenter and President, NCUGF Rural Service Club), 2024
- **High Distinction and Public Service Scholarship**, National Central University (as President, NCUGF), 2024

Teaching Experience

National Central University

Feb – Jun 2025

Teaching Assistant, Courses: *Natural Language Processing* and *Enterprise Application of Pattern Recognition*

- Assisted in lecture preparation and grading for NLP topics for **60** students and image recognition topics for **60** students, analyzing the underlying AI technologies of real-world company products to support students in final project research.

Work Experience

Chunghwa Telecom

AI Software Development Intern

Sep – Dec 2025

Taipei, Taiwan

Tools: n8n workflow, Google Gemini Models, Azure OpenAI API, React, Node.js, Python

- Spearheaded the full-stack development of an intelligent project management system, spanning UI/UX design and n8n workflow backend orchestration, driven by RAG-based agentic models to enhance operational efficiency by **33%**.
- Selected to represent the department at the *2025 IEEE Global Communications Conference*, introducing the department's AI-powered intelligent customer service products to international industry and academic professionals.

Industrial Technology Research Institute

AI Software Summer Intern

Jun – Sep 2024

Hsinchu, Taiwan

Tools: Raspberry Pi, Linux, Azure OpenAI API, Flask, Node.js, Git, Python

- Built and deployed an LLM-based service chatbot at *Taipei Songshan Sports Center*, integrating Azure OpenAI with Linux backend services to provide **24/7** automated customer support, reducing customer queue time by **50%** and improving service efficiency.
- Selected to demonstrate the department's AI chatbot at the *2024 Smart Healthcare Summit* and *2024 Healthy Ageing Tech Show*.

Skills

Programming: Python (PyTorch), C/C++, JavaScript (Node.js, Express, React)

Systems & Deployment: Linux, Docker, Azure, REST APIs, Git, OpenAI API

Machine Learning: DRL (DDPG, PPO, Meta-RL), Generative Models (VAE, Diffusion), GNN, Transformer, LLMs

AI Applications: RAG pipeline, Agentic workflow (n8n), LLM-based chatbot, Full-stack AI integration

Languages: English (TOEFL 89 / TOEIC 810), Mandarin Chinese (Native)

Leadership & Extracurriculars

NCUgF Rural Service Club

Feb 2023 – Jul 2025

Club President & Service Team's Group Leader

- Secured multiple awards and scholarships for the club through service initiatives as club president (see Honors & Awards).
- Led and coordinated educational camps for **160** children across remote elementary schools in Yilan in February 2023 and Taichung in January 2024, designing interactive lessons and managing team activities to foster creativity and learning.

NCU Wireless Communication and Network Multimedia Camp

Jul 2023

AI Instructor & Course Team Member

- Created AI course materials and delivered lectures on machine learning and LLM fundamentals to **70** high school students.
- Guided students in completing hands-on Arduino projects to foster practical understanding of wireless communication.